G1 Datasets (Small datasets)

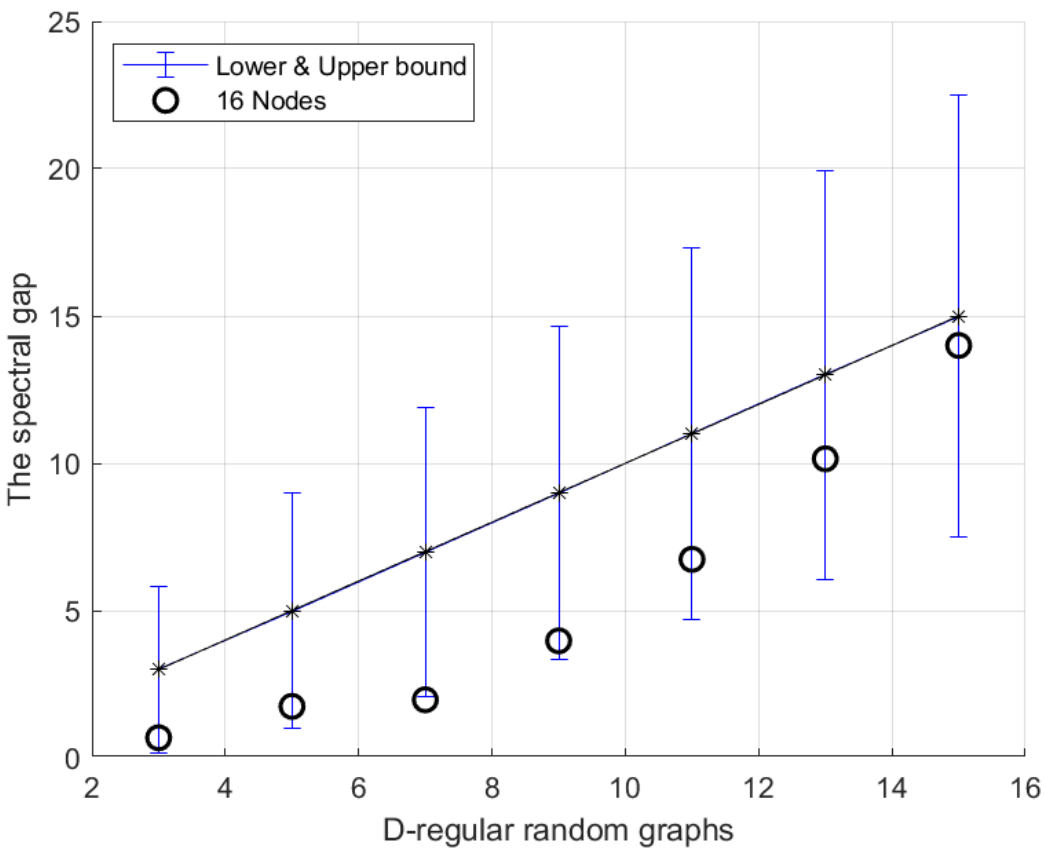
Datasets	Training Points	Testing Points	Features
ionosphere	300	51	34
svmguid	3089	4000	4

d-regular expander graphs with low degrees designed for training these small datasets with 16 nodes

where d = {3,5,7,9,11,13,15}

Results

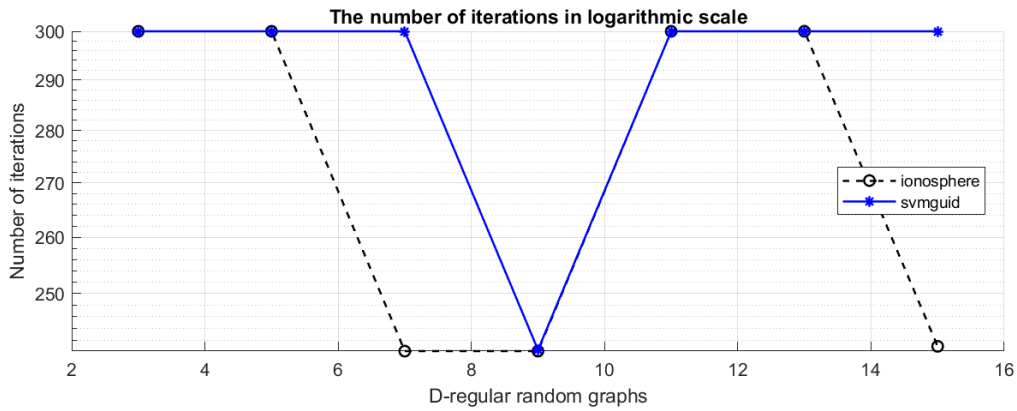
1. The upper and lower bounds of the spectral gap for random d-regular expander graphs



This figure illustrates the upper and lower bounds of the spectral gap (λ2) for random d-regular expander graphs with 16 nodes, where d ranges from 3 to 15.
The vertical bars show the theoretical range of λ2 , while the plotted points represent the measured spectral gaps of the constructed graphs.

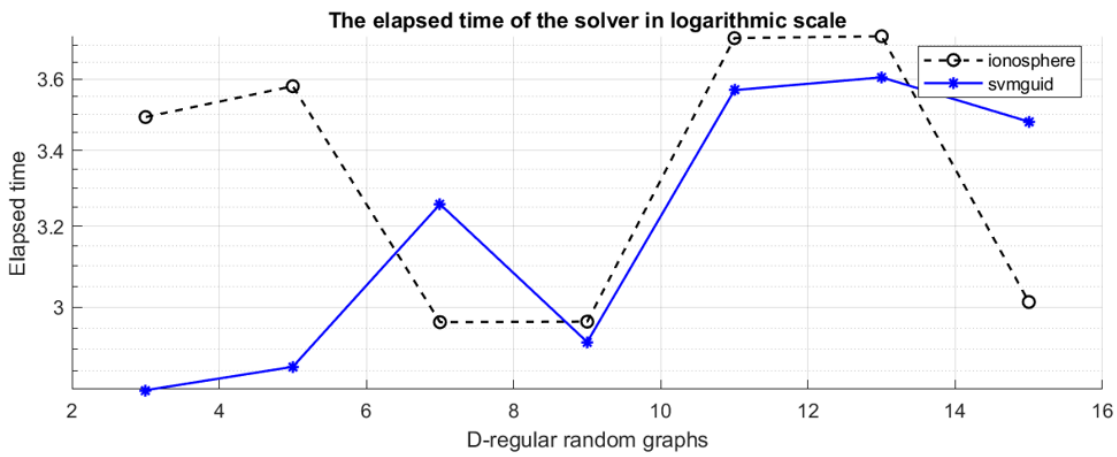
Values marked with * correspond to Ramanujan graphs, which achieve spectral gaps close to the upper bound and reflect their superior connectivity.
Values marked with o correspond to random d-regular graphs, which lie within the theoretical bounds but can vary due to randomness in construction.
As the degree increases, the spectral gap generally grows, meaning the graph becomes more connected and supports faster consensus.
This figure confirms that the generated graphs follow the expected theoretical behaviour and are suitable for analysing ADMM convergence performance.

2. The number of iterations in logarithmic scale



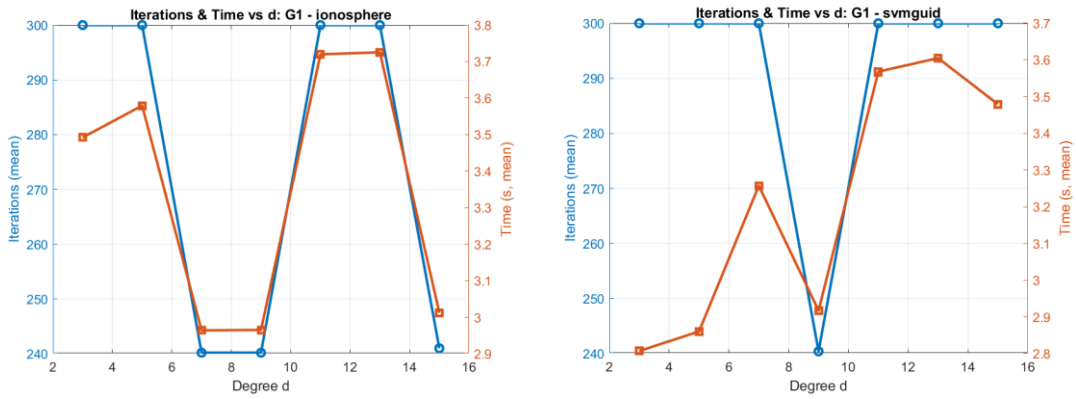
- When λ2 is larger, information spreads faster through the network, and consensus is achieved in fewer iterations.
- When λ2 is small, the consensus is slow, and the solver keeps iterating until it reaches the iteration cap.

3. The elapsed time of the solver in logarithmic scale



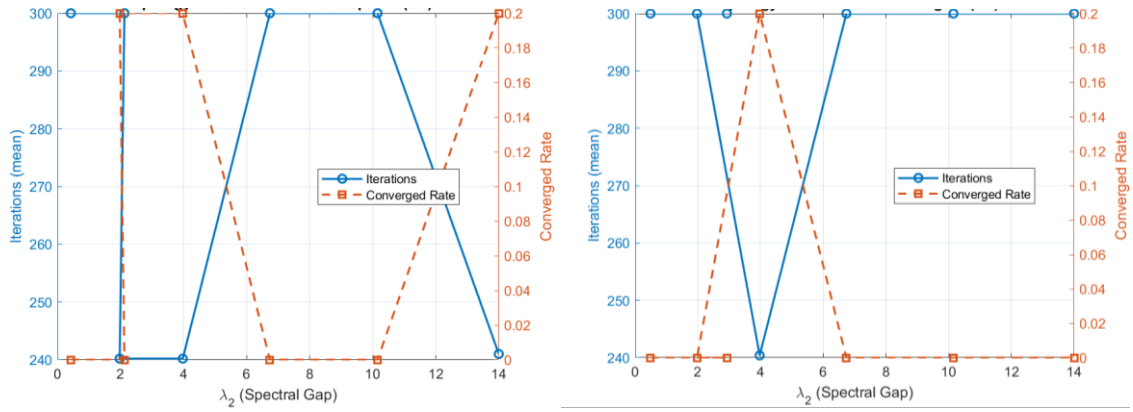
- **The elapsed time is shaped by a trade-off:** sparse graphs (low d) take long due to poor connectivity, while dense graphs (high d) incur higher communication cost. The fastest runtimes occur at **moderate degrees** (7–9), where spectral gap is large enough to speed convergence but communication overhead remains small.

4.iterations & time vs degree : ionosphere and svmguid (G1)



- Illustrating that solver performance is governed by the **spectral gap of the graph**: better connectivity reduces iterations and time, but because degree is only an indirect measure, the actual runtime shows fluctuations instead of a smooth trend.

5. Topology effect on ADMM: ionosphere and svmguid datasets (G1)



- This plot shows that ADMM's efficiency depends directly on the **spectral gap** of the graph. A larger λ2 means faster consensus (fewer iterations) and higher convergence rates, while a smaller λ2 leads to slow or failed convergence. Performance tracks λ2, not degree, which is why the curves are not monotonic.

G2 Datasets (Large Datasets)

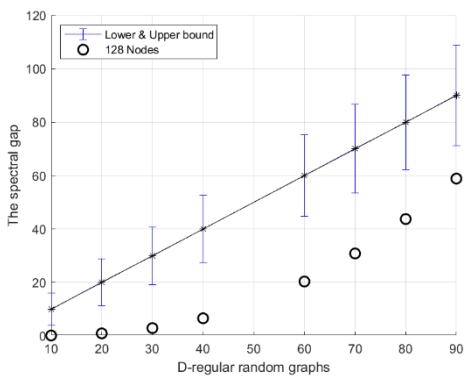
Datasets	Training Points	Testing Points	Features
phishing	11055	1655	68
a9a	32561	16281	123
ijcnn	35000	14990	22
skinNonskin	37492	5000	3

d-regular expander graphs with higher degrees designed for training these large datasets with 128 nodes

where d = {10,20,30,40,60,80,90}

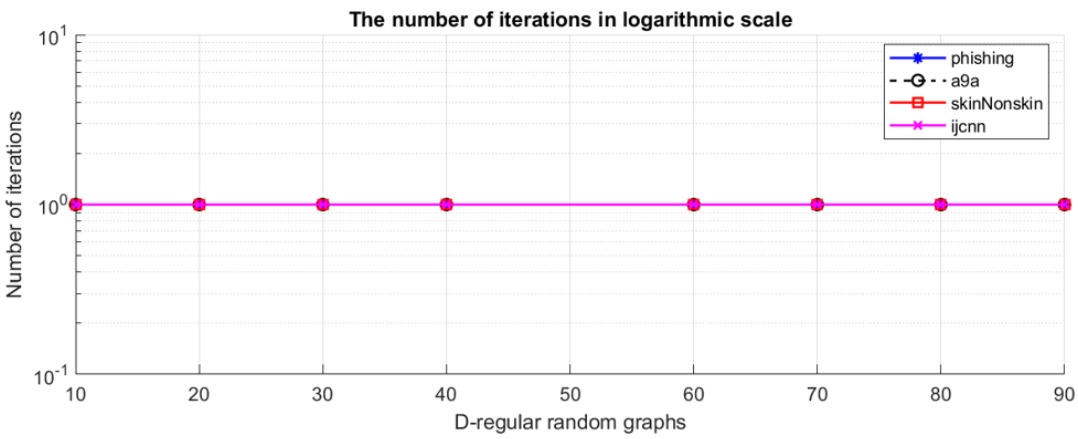
Results

1. The upper and lower bounds of the spectral gap for random d-regular expander graphs



This figure shows the upper and lower bounds of the spectral gap (λ2) for random d-regular and Ramanujan graphs with 128 nodes, where d = {10, 20, 30, 40, 60, 80, 90}.
The vertical bars represent the theoretical range of λ2 while the plotted points indicate the actual spectral gaps measured from the constructed graphs.
The Ramanujan graph values (marked with *) lie close to the upper bound, reflecting their near-optimal connectivity.
The random d-regular graph values (marked with o) fall within the theoretical bounds but may vary depending on the specific graph instance.
As the degree increases, the spectral gap generally becomes larger, indicating better connectivity and faster potential convergence in consensus algorithms.
Overall, the figure confirms that the graphs used in the experiments respect the theoretical spectral gap limits and validates their suitability for evaluating ADMM performance.

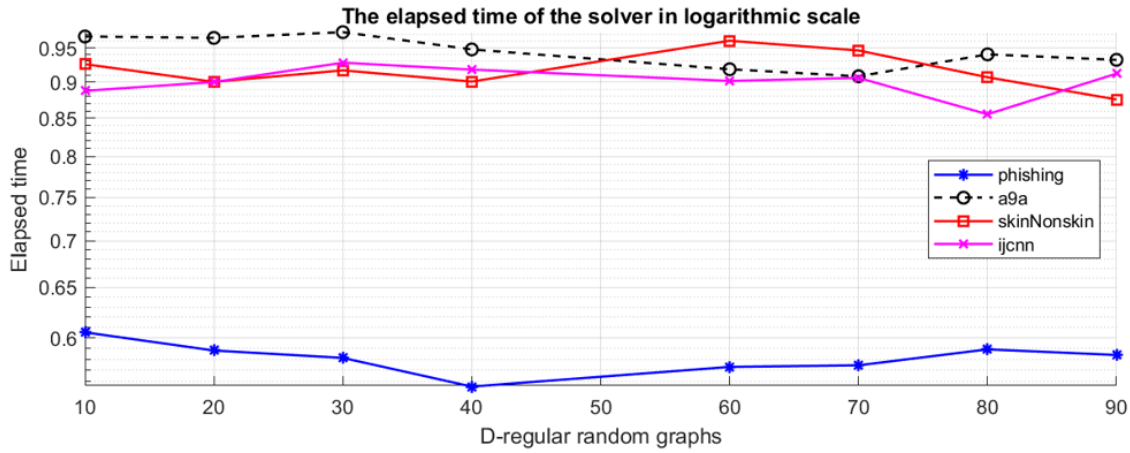
2. The number of iterations in logarithmic scale



This figure shows how the number of ADMM iterations varies with the degree of 128-node d-regular graphs for the G2 datasets (phishing, a9a, skinNonskin, and ijcnn).
Across all datasets and degrees, the number of iterations remains essentially constant at one, indicating that consensus is reached almost immediately.
This behaviour demonstrates that for large datasets, the solver is not sensitive to graph connectivity once a basic level of communication is present.
The spectral gap provided by even modest degrees is sufficient to ensure immediate convergence.

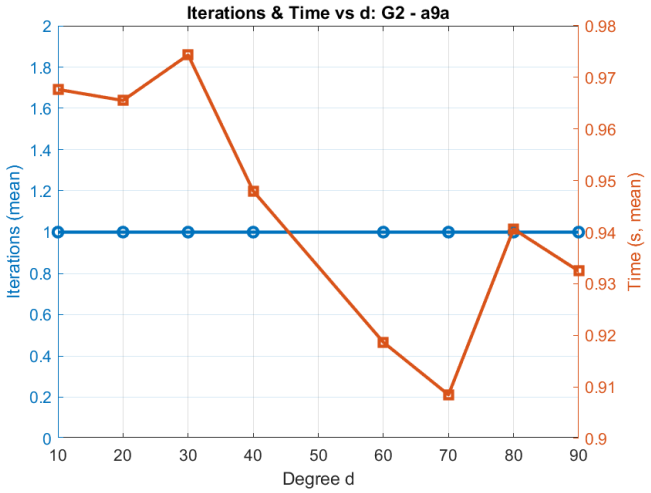
As a result, the iteration curves remain flat, and the graph highlights that topology has little to no effect on the iteration requirements for these datasets. This confirms that ADMM performance in G2 is dominated by dataset size and conditioning, not by the degree of the communication graph.

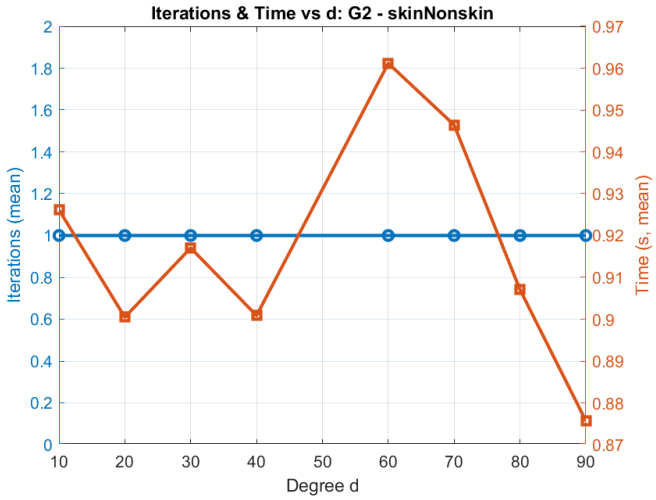
3. The elapsed time of the solver in logarithmic scale

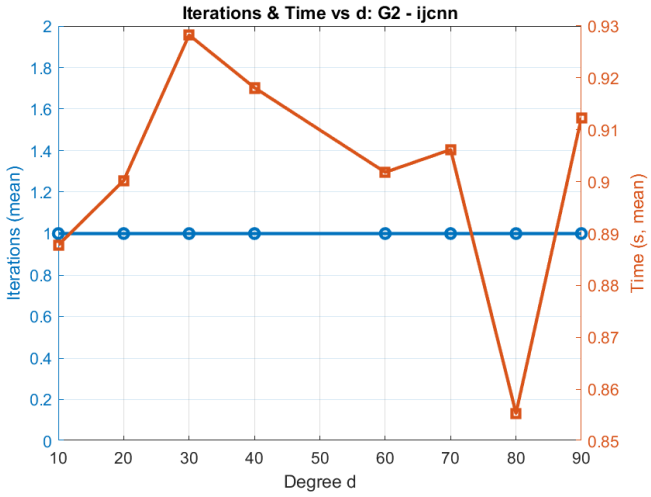


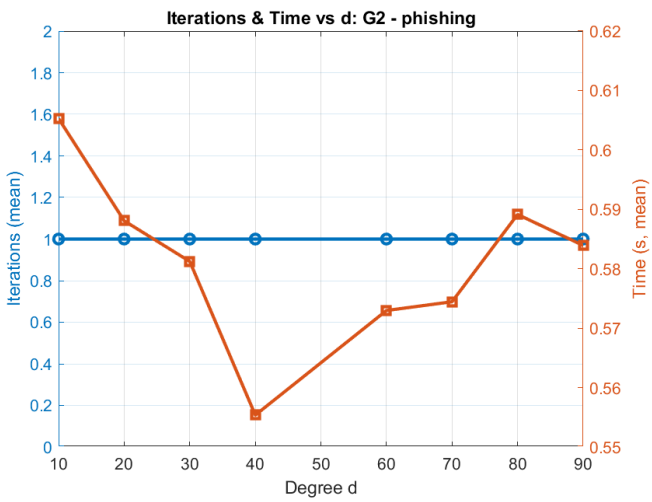
This figure illustrates the average elapsed runtime of the solver as the degree of 128-node d-regular graphs increases for the G2 datasets.
Since the number of iterations remains fixed at one, elapsed time does not vary significantly across degrees.
Small fluctuations that appear in the curves are caused by changes in communication overhead: as the degree grows, each iteration involves more neighbour exchanges, slightly increasing the per-iteration cost.
At moderate degrees, runtime tends to be marginally lower because communication remains efficient while connectivity is already sufficient.
At very high degrees, the cost per iteration rises slightly, leading to minor increases in runtime despite convergence being immediate.
Overall, this figure conveys that solver runtime for G2 datasets is nearly constant and that graph topology plays only a minimal role once datasets are large and well-conditioned.

4. Iterations & Time vs d: G2 Datasets



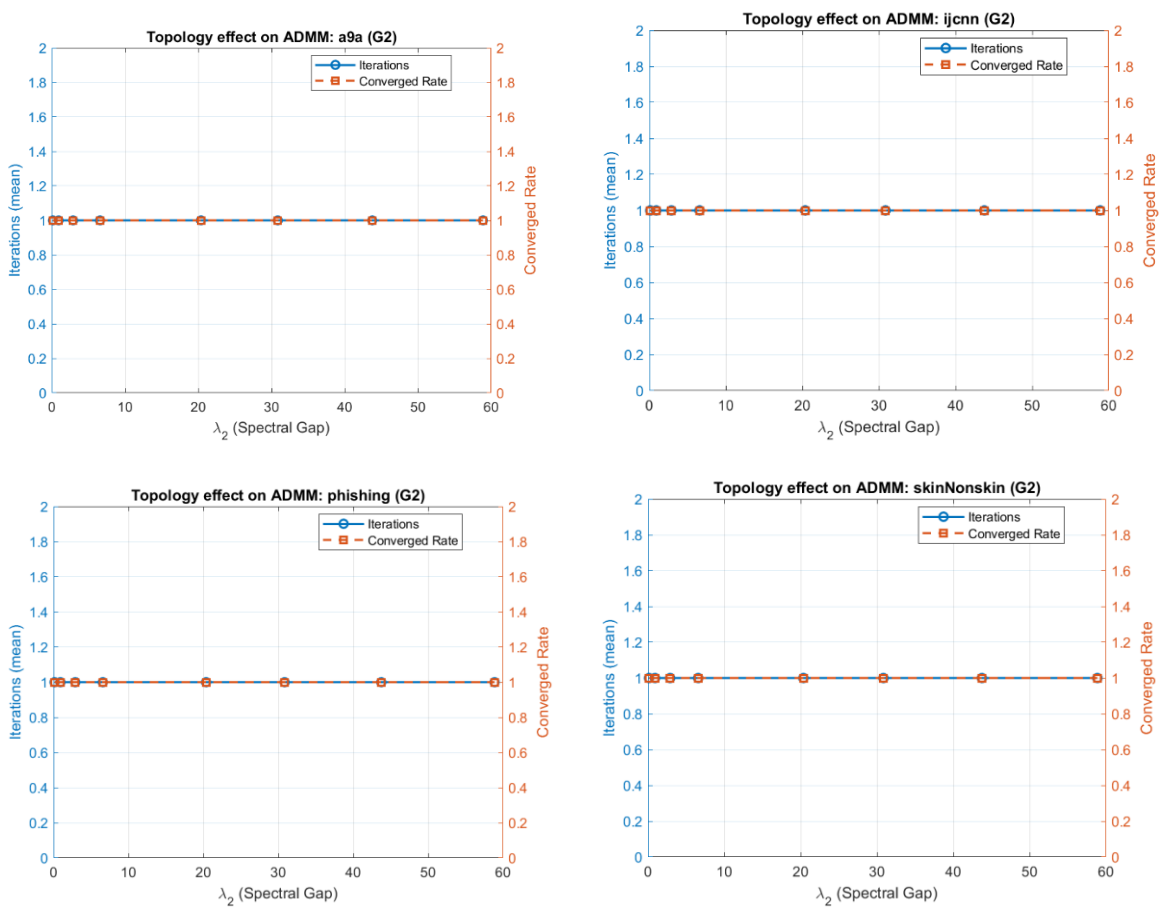






For large G2 datasets (a9a, ijcnn, phishing, skinNonskin), ADMM converges in a single step across all graph degrees. The runtime differences are marginal and only reflect per-iteration communication cost. This highlights that for big, well-conditioned problems, connectivity is sufficient even at low degrees, and topology has little effect on solver efficiency.

5. Topology effect on ADMM: G2 Datasets



For the G2 datasets, the topology effect plots indicate that the number of iterations remains constant at one across all spectral gap values.
The convergence rate stays at 100%, showing that the solver always reaches consensus successfully.
This demonstrates that spectral gap does not influence the convergence behaviour in these cases.
The large size of the datasets makes local models already well aligned before communication begins.
As a result, the ADMM algorithm converges in a single step, regardless of graph structure.
In contrast to G1 datasets, solver performance in G2 is independent of topology and consistently stable.